

PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

PROJECT SYSTEM FLOW PIPELINE (DFD LEVEL 0 & 1):

[Client Browser View Node / Form Inputs Grid (N, P, K, Temp, Humid, pH, Rain)]

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▼ (POST Request Matrix Parameters Data Pipeline)

[Flask Application Core Backend Processor (app.py Web Service)]

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|—► Reads Historical Matrices —► [Local Storage: data/crop_data.csv]

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▼ (Vector Array Real-Time Feature Mapping)

[Sklearn Loaded Ensemble Inference Classifier Engine (models/crop_model.pkl)]

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▼ (Target Class Category Output Label Identification Match)

[Process Result Evaluation Box Rendering Engine (index.html Response Alert Box View)]